

Introduction

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Introduction

What conveys the study of econometrics?

The unification of three main areas:

1. Statistics
2. Economic theory
3. Mathematics

“Econometrics conveys application of mathematical statistics and the tools of statistical inference to the empirical measurement of relationships postulated by an underlying theory.”

Why Econometrics?

- Econometrics is not just math + stats + economics
- It is about answering **real questions**:
 - Does raising the minimum wage reduce employment?
 - What is the return to education?
 - Do social programs reduce poverty?
- It provides a **bridge between theory and data**

A Brief History

- **1930s:** Creation of the Econometric Society
- **1944:** Trygve Haavelmo's "Probability Approach in Econometrics"
- Econometrics as the *science of testing economic theory with data*
- Quote: "All models are wrong, but some are useful" – George Box

The practice of econometrics

Theoretical econometrics vs. applied econometrics

This course is designed for the applied economist.

Applied econometric methods will be used for estimation of important quantities, analysis of economic outcomes such as policy changes, markets or individual behavior, testing theories, and for forecasting.

Econometric Modeling

Consider the Keynesian consumption function: $C = f(X)$, where X is disposable income.

Theoretical postulates: dC/dX is positive and $0 < dC/dX < 1$.

Implications:

1. $MPC > 0$
2. $APC = C/X$ falls with income

Keynes' Consumption Function

How do we test this theory?

Estimate the equation $C = \alpha + \beta X$ and test whether $\alpha > 0$ and $0 < \beta < 1$.

Some points to note:

1. Notice the behavioral aspect of the model.
2. The model states an unambiguous and deterministic relationship between the dependent and independent variables.
3. Models are only simplifications of reality; we need a stochastic element.
4. A probabilistic model is less precise but more robust!

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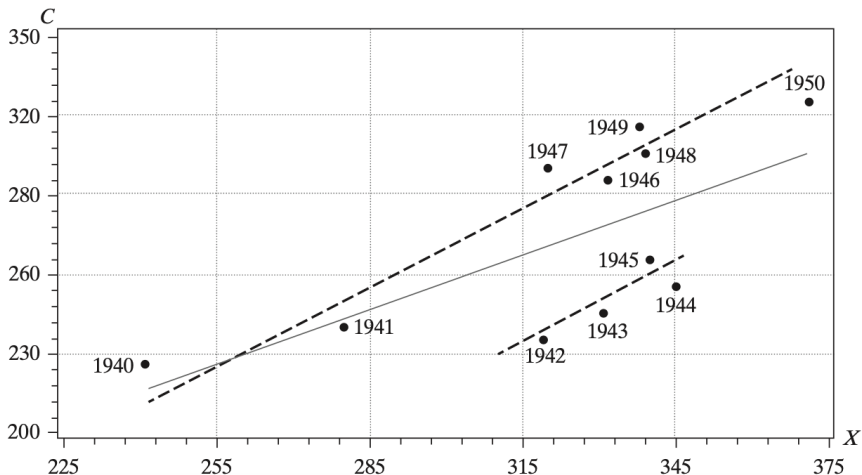
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From the Model to the Real World

FIGURE 2.1 Consumption Data, 1940–1950.



From the Model to the Real World

The next step is to include a random component: $C = F(X, \varepsilon)$

We usually assume an additive relationship.

We also need to consider the wartime period into the empirical model.

Here's an empirical model:

$$C = \alpha + \beta X + d_{waryears} \delta_W + \varepsilon$$

Another Example

How would you model the function $\text{earnings} = f(\text{education})$?

Famous Empirical Applications

- **Labor Economics:** Minimum wage and employment
 - Card & Krueger (1994): natural experiment in fast food restaurants
- **Education:** Returns to schooling
 - Mincer (1974), Angrist & Krueger (1991)
- **Development:** Impact of conditional cash transfers
- **Macroeconomics:** Forecasting inflation and GDP growth

Beyond Correlation: Causality

- Econometrics is about more than prediction
- Key distinction: **Correlation vs. Causation**
- Identification problems:
 - Omitted variables
 - Reverse causality
 - Measurement error
- Econometrics provides tools to address them

New Frontiers

- **Big Data** and administrative microdata
- Integration with **Machine Learning**
- Structural vs. reduced-form econometrics
- Opportunities: richer datasets, more complex models
- Challenge: theory still matters!

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